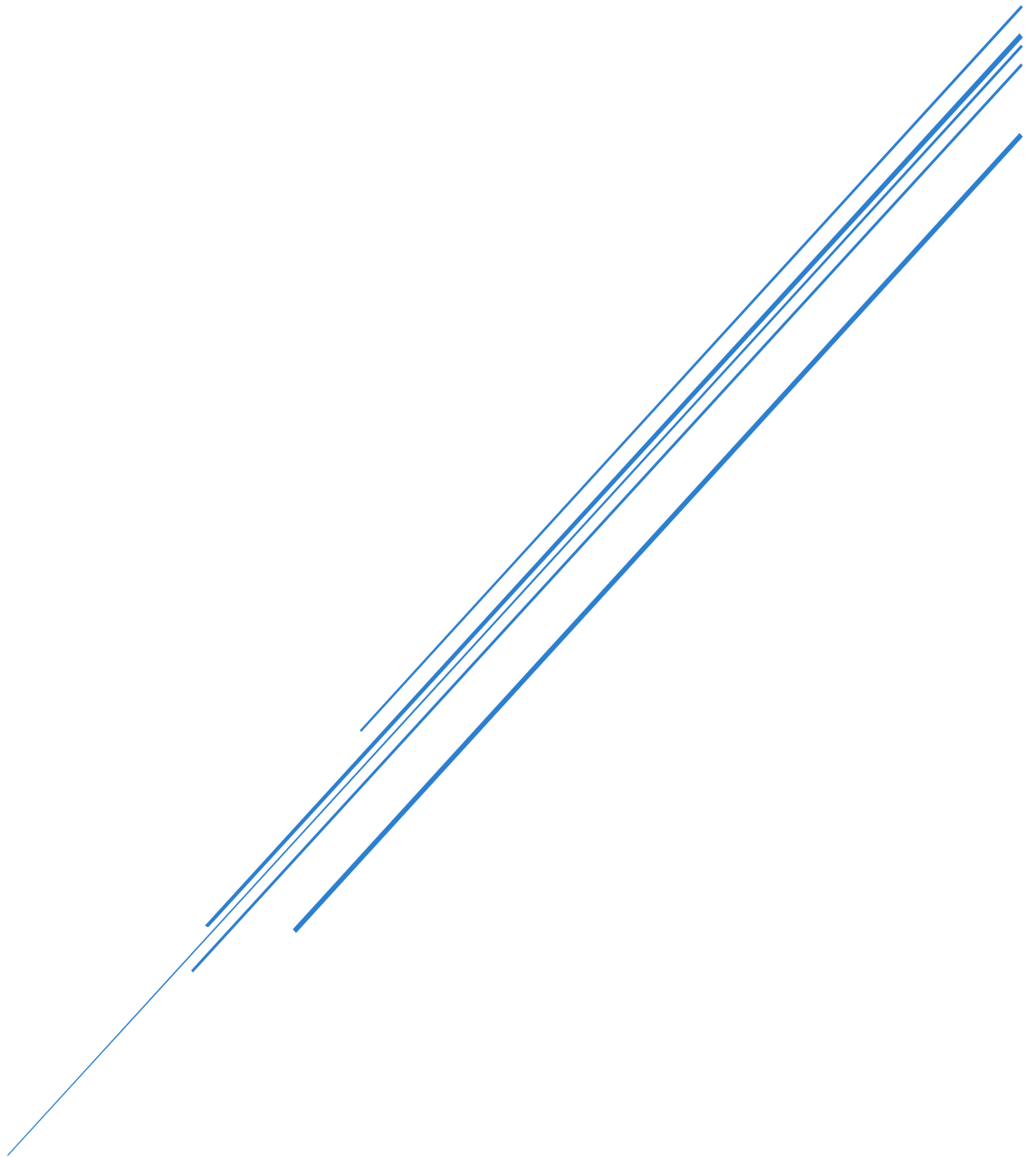


SYNAPSE STOCKS

AI-POWERED STOCK INTELLIGENCE PLATFORM



PAWGRAMMERS TEAM
Vibe Coding Project

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

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SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

Welcome to SynapseStocks

A full-stack, agentic AI platform built to help investors and analysts decode the stock market with clarity and confidence.

Whether you're tracking financial trends, climate impact, or social sentiment, SynapseStocks empowers you with explainable, multi-factor insights through:

- Modular AI agents tailored to different sectors
- A sleek, responsive dashboard built with React and Material-UI
- Secure JWT-based authentication
- A clean monorepo structure for easy development and deployment

This project is ideal for developers, data scientists, and financial analysts looking to explore intelligent market analysis tools in a demo or educational setting.

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

Project Details

Project Name: SynapseStocks

Category: Full-stack AI-powered financial analytics platform

Purpose: SynapseStocks is built to help investors and analysts make smarter, more informed trading decisions. By combining explainable AI with multi-factor market insights, it turns complex data into actionable intelligence.

Technical Overview

Here's a quick look at the tech powering SynapseStocks:

- **Frontend:** Built with React and styled using Material-UI for a clean, responsive interface
- **Backend:** Node.js with Express and lowdb for lightweight, fast data handling
- **Authentication:** Secure login system using JWT tokens
- **Architecture:** Organized as a monorepo with separate client/ and server/ directories for clarity and scalability
- **AI Agents:** Modular and sector-aware agents that analyze different market signals

Key Features

- A dynamic dashboard with agent cards that deliver insights on:
 - News
 - Financials
 - Chemical Prices
 - Climate Impact
 - Social Sentiment
 - Regulatory Factors

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- Easily extendable backend agents—add new ones or customize existing ones with minimal effort
- Environment-aware API configuration for smooth deployment across different stages
- A modern, intuitive UI that makes navigating insights effortless
- Secure user sessions powered by JWT-based authentication

How to Get Started

Setting up SynapseStocks is simple:

1. Clone the repository
`git clone https://github.com/Cognizant-SPE/vibe-coding-synapsestocks.git`
2. Set up the backend
`cd server && npm install && cp .env.sample .env && npm run dev`
3. Set up the frontend
`cd ../client && npm install && cp .env.sample .env && npm start`

Innovation

SynapseStocks brings a fresh perspective to financial analytics by combining modular AI with a user-first design and secure infrastructure. Here's what makes it stand out:

Smarter AI Architecture

- Each market factor is handled by a dedicated AI agent, offering focused and explainable insights.
- The modular design means agents can be added, removed, or updated without disrupting the system—perfect for evolving market needs.

Sector-Specific Intelligence

- Agents are tuned to specific domains like climate, chemicals, and regulatory trends, giving users deeper and more relevant analysis than generic tools.

PROJECT DETAILS AND INNOVATION

SYNAPSE STOCKS: AI-POWERED STOCK INTELLIGENCE PLATFORM

Built for Developers

- The monorepo structure makes development and deployment straightforward.
- The lightweight backend using lowdb is ideal for quick prototyping and educational use.

Security You Can Trust

- JWT-based authentication ensures that user data and insights are protected.
- Environment-aware API setup makes it easy to switch between development and production environments.

Designed for Users

- The dashboard, built with Material-UI, offers a clean and modern interface.
- Agent cards simplify complex data, making it easy to understand and act on insights.